Can sector correlations improve weekly ETF rankings?

The first historical run produced near-zero ranking accuracy for both models. Whether the correct correlation weights add predictive value remains open.

-0.004	-0.011	+0.007
GATv2 mean test IC	MLP mean test IC	Difference in mean IC

What the team measured

Every five trading days, each model ranked eligible sector ETFs by their forward five-bar return, measured from the next open. Both received 20- and 60-day momentum and 20-day volatility, ranked across ETFs. The GATv2 also connected every ETF pair, weighting each link by its absolute 60-day return correlation. The MLP scored each ETF independently.

Rank IC measures agreement between predicted and realized return rankings. The reported means cover 10 walk-forward test folds ending before 2024. Without results for individual folds, the 0.007 gap gives no measure of consistency or uncertainty.

Why the first result is inconclusive

The team observed nearly identical ETF scores. On synthetic data with a known signal, a higher learning rate helped training move beyond that flat start. The revised setup uses a 0.01 learning rate, 20-epoch patience and one graph layer. No newer ETF run was available for this review.

Proposed experiment

1. Check the revised baseline. Compare GATv2 and MLP validation rank IC by fold and inspect the spread of scores across ETFs on each ranking date. Distinct scores show that the model separates sectors. Positive rank IC shows whether the ordering is useful.

2. Isolate the value of the weights. Retrain the same GATv2 after permuting ETF identities on the weighted graph while keeping features and labels attached to their original ETFs. This preserves the complete graph and its weight distribution while changing which pairs receive the strongest weights.

3. Compare matched runs. Record real minus shuffled validation IC for each fold and training seed, repeat across several shuffles, and report the mean difference and how often it is positive. A consistent gain would support using the actual ETF pairings. Reserve the three holdout folds starting in 2024 for the final evaluation.

Sources: Team-provided PLAN.md, config.yaml and sector_etf_corr.yaml.